1.0 GENERAL

1.1 ALL WORKS SHALL BE UNDERTAKEN IN ACCORDANCE WITH THE LATEST EMPLOYER APPROVED, SIGNED AND STAMPED EDITIONS OF RELEVANT STANDARDS AND THEIR AMENDMENTS, PROJECT SPECIFICATIONS AND THE DESIGN AND STANDARD DRAWINGS AS SPECIFIED.

1.2 DESIGN DRAWINGS SHALL NOT BE SCALED AND ALL DIMENSIONS SHALL BE READ DIRECTLY FROM DESIGN DRAWINGS. ANY DISCREPANCIES WITHIN THE DOCUMENTATION SHALL BE REFERRED TO THE EMPLOYER FOR CLARIFICATION BEFORE PROCEEDING.

1.3 PRODUCTS EQUIVALENT TO THOSE SPECIFIED ON THE DESIGN DRAWINGS MAY BE USED ONLY WITH APPROVAL OF THE EMPLOYER.

1.4 FOR PROJECT SPECIFIC NOTES REFER TO DESIGN DRAWINGS.

1.5 ALL DIMENSIONS ARE GIVEN IN MILLIMETRES UNLESS NOTED OTHERWISE.

1.6 ALL LEVELS ARE GIVEN IN METRES UNLESS NOTED OTHERWISE.

1.7 VERTICAL DATUM (TO BE SPECIFIED)

1.8 HORIZONTAL GRID (TO BE SPECIFIED.)

2.0 EXCAVATION AND FOUNDATIONS

2.1 EXCAVATION AND FOUNDATION PREPARATION SHALL COMPLY WITH STANDARD SPECIFICATION MD-101-SMEM-EG-CV-SPC-0001.

2.2 NO BACKFILLING SHALL COMMENCE AGAINST WALLS UNTIL 28 DAYS HAVE ELAPSED FROM COMPLETION OF THE CONCRETE POUR OR APPROVED PROPPING HAS BEEN INSTALLED.

2.3 ANY OVER EXCAVATION AT FOOTING LOCATIONS SHALL BE MADE GOOD WITH SELECT FILL COMPACTED TO AT LEAST THE BEARING CAPACITY OF THE SURROUNDING UNDISTURBED SOIL IN ACCORDANCE WITH STANDARD SPECIFICATION MD-101-SMEM-EG-CV-SPC-0001. ALTERNATIVELY LEAN MIX CONCRETE MAY BE USED.

2.4 SOIL UNDER SLABS ON GRADE SHALL BE COMPACTED TO A MINIMUM DEPTH OF 300 mm TO ACHIEVE 95% MODIFIED MAXIMUM DRY DENSITY AS MEASURED BY ASTM D1557 FOR SANDS AND LABORATORY TEST ASTM D2922 FOR COHESIVE SOIL.

2.5 SOIL UNDER FLAT BOTTOMED TANKS SHALL BE COMPACTED AS PER NOTE 2.4 TO ACHIEVE 98% MODIFIED MAXIMUM DRY DENSITY.

2.6 ALL EXCAVATIONS FOR FOOTINGS SHALL HAVE A MINIMUM THICKNESS OF 75mm OF LEAN MIX CONCRETE (GRADE IN ACCORDANCE WITH STANDARD SPECIFICATION) APPLIED IMMEDIATELY ON COMPLETION OF THE EXCAVATION. NO EXCAVATION SHALL BE LEFT OPEN OVERNIGHT WITHOUT POURING LEAN MIX CONCRETE.

2.7 TO FACILITATE DRAINAGE AWAY FROM CONCRETE STRUCTURES THE CONCRETE CONTRACTOR SHALL BACKFILL AND COMPACT LOCALLY BEHIND ALL KERBS, AND AROUND THE EDGES OF ALL SLABS AND ISOLATED PEDESTALS, TO THE DETAILS SHOWN ON THE DESIGN DRAWINGS AND AS DIRECTED BY THE EMPLOYER.

3.0 FORMWORK

3.1 ALL FORMWORK AND FALSEWORK SHALL BE IN ACCORDANCE WITH STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0006 AND SHALL COMPLY WITH ACI 347.

3.2 ALL EXPOSED EDGES OF CONCRETE SHALL BE PROVIDED WITH A 20mm x 20mm CHAMFER OR FILLET.

4.0 CONCRETE

4.1 ALL CONCRETE WORKS SHALL BE IN ACCORDANCE WITH STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0005 AND 0006 AND ACI 301.

4.2 CONCRETE FINISHES SHALL COMPLY WITH ACI 302.1R, STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0006 AND AS FOLLOWS, UNLESS NOTED OTHERWISE ON THE DESIGN DRAWINGS:

THE CONCRETE STANDARD NOTES SHOWN ON THIS DRAWING APPLY UNLESS NOTED OTHERWISE ON THESE STANDARD DRAWINGS OR ON THE DESIGN DRAWINGS

4.3 CONCRETE FINISHES (REFERENCE TO MD-101-SMEM-EG-ST-SPC-0005)

FINISHING UNFORMED SURFACES	
FINISH	TYPICAL APPLICATION
SCRATCHED	SURFACE OF ROADS, FOUNDATIONS, BEDS, SLABS AND STRUCTURAL MEMBERS TO BE COVERED BY BACKFILL, SUBSEQUENT STAGES OF CONSTRUCTION, BONDED CONCRETE, TOPPING OR CEMENT MORTAR BEDS TO RECEIVE PAVING AND EXPOSED SURFACES WHERE SPECIFIED
WOOD FLOATED	FOR WALKS, DRIVES, STEPS, RAMPS AND FOR SURFACES TO RECEIVE WATERPROOFING, INSULATION, MASTIC OR BLOCK OR TILE PAVING
STEEL TROWEL	FOR FLOORS, PAVING, TOPS OF WALLS, COPINGS OR OTHER MEMBERS EXPOSED TO WEATHERING, SURFACE BEDS AND SLABS TO RECEIVE THIN FLEXIBLE SHEET AND/OR TILE
BROOM	SURFACES WHICH ARE FOR FOOT TRAFFIC

FINISHING FORMED SURFACES	
FINISH	TYPICAL APPLICATION
ROUGH-FORM	SURFACES AGAINST BACKFILL OR AGAINST WHICH FURTHER CONCRETE WILL BE PLACED
SMOOTH-FORM	SURFACES WHICH ARE PERMANENTLY EXPOSED TO VIEW, BUT WHERE THE HIGHEST STANDARD OF FINISH IS NOT REQUIRED
SMOOTH-RUBBED	SURFACES PROMINENTLY EXPOSED TO VIEW WHERE GOOD APPEARANCE AND ALIGNMENT ARE OF SPECIAL ARCHITECTURAL IMPORTANCE

4.4 CONCRETE CURING SHALL COMPLY WITH STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0006.

4.5 FOR CONCRETE STRENGTH GRADES REFER STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0005. DESIGNER SHALL SPECIFY CONCRETE GRADE ON DESIGN DRAWING.

4.6 ADDITIVES SHALL NOT BE USED WITHOUT PRIOR APPROVAL OF THE EMPLOYER.

4.7 THE MAXIMUM FREE FALL OF CONCRETE DURING PLACEMENT SHALL NOT EXCEED 2 METRES OR AS SPECIFIED IN STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0006.

4.8 CONSTRUCTION JOINTS AND SLEEVES SHALL NOT BE USED IN POSITIONS OTHER THAN THOSE INDICATED ON THE DESIGN DRAWINGS WITHOUT PRIOR APPROVAL OF THE EMPLOYER.

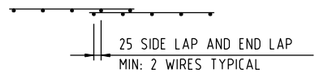
4.9 A VAPOUR BARRIER MEMBRANE CONSISTING OF A 0.25mm POLYTHENE FILM SHALL BE PLACED UNDER ALL SLABS POURED ON GROUND.
LAP VAPOUR BARRIER MEMBRANE MINIMUM 100mm AND SEAL WITH ADHESIVE TAPE TO ENSURE A COMPLETE MOISTURE BARRIER.

5.0 REINFORCEMENT

5.1 REINFORCING STEEL BAR SHALL BE DEFORMED BAR CONFORMING TO MP-101-SMEM-EG-ST-SPC-0006 AND ASTM A615/A615M, GRADE 60 UNLESS NOTED OTHERWISE.

5.2 WELDED STEEL FABRIC SHALL CONFORM TO A185/A185M AND A884/A884M.

ALL FABRIC SHALL BE SUPPLIED IN FLAT SHEETS. LAP FABRIC AS SHOWN BELOW.



5.3 SPLICES IN REINFORCEMENT SHALL BE PROVIDED IN THOSE POSITIONS SHOWN ON THE DRAWINGS. IN ALL CONCRETE ELEMENTS WITH LONG CONTINUOUS BARS, SPLICES SHALL BE STAGGERED IN ADJACENT BARS (START TO START) BY 48xBAR DIAMETER MINIMUM. SPLICES SHALL NOT BE PROVIDED WHERE CRITICAL BENDING STRESSES ARE DEVELOPED

5.4 MINIMUM LAP OR EMBEDMENT LENGTHS FOR REINFORCEMENT SHALL BE 40xBAR DIAMETER. MINIMUM LAP LENGTH SHALL SHALL BE IN ACCORDANCE WITH ACI 318. WHEN LAPPING BARS OF DIFFERENT SIZE THE LAP LENGTH FOR THE SMALLER BAR SHALL APPLY.

5.5 EXPLANATION OF NOTATION FOR REINFORCEMENT USED ON DRAWINGS:
EXAMPLE: 16-D24(002)-300B1

16 NUMBER OF REINFORCING BARS
D24 BAR GRADE 'D' (DEFORMED BAR) AND DIAMETER '24'
(002) BAR MARK NUMBER (WHEN SPECIFIED)
300 DISTANCE BETWEEN BAR CENTRES
B1 DESIGNATION OF REINFORCEMENT LAYER WHENEVER SPECIFIED (REFER DIAGRAM 1)

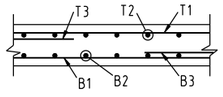


DIAGRAM 1 - CONCRETE SLAB

T -TOP
B -BOTTOM
NF -NEAR FACE
FF -FAR FACE
EW-EACH WAY
EF -EACH FACE

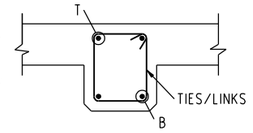


DIAGRAM 2 - CONCRETE BEAM

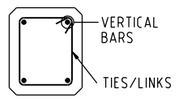


DIAGRAM 3 - CONCRETE COLUMN

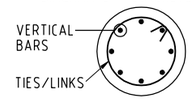


DIAGRAM 4 - CONCRETE COLUMN

5.6 MINIMUM CONCRETE PROTECTIVE COVER FOR REINFORCEMENT SHALL BE IN ACCORDANCE WITH SBC 304 OR THE FOLLOWING WHICHEVER IS GREATER:

No.	CONCRETE EXPOSURE CONDITION	CONCRETE COVER (mm)	REMARKS
1	EXPOSED TO SEA WATER OR SOIL		
(a)	UNDERGROUND FOUNDATION, TIE-BEAMS, WALLS AND UNDERSIDE/ SIDE OF SLABS.	75	WITH SURFACE PROTECTION
(b)	TOP SIDE OF SLABS	50	WITH FINISHED FORMED SURFACES
2	EXPOSED TO WEATHER		
(a)	UNDERGROUND & ABOVE GROUND FOUNDATIONS	75	WITH SURFACE PROTECTION
(b)	COLUMNS, WALLS AND BEAMS	60	FOR STRUCTURE & BUILDING ABOVE GROUND
(c)	JOIST AND SLABS	50	SMALL BEAMS & BUILDING SLAB (OUTDOOR)
(d)	ALL WELDED WIRE FABRIC	50	FOR 100mm THICK PAVEMENT
3	NOT EXPOSED TO WEATHER OR SOIL		
(a)	OUTDOOR UNPROTECTED FACES	50	WITH FINISHED FORMED SURFACES
(b)	ALL INTERNAL PROTECTED SURFACES	40	BUILDING SLAB (GROUND FLOOR)
(c)	ALL WELDED WIRE FABRIC	25	BUILDING SLAB (ABOVE GROUND)
4	PRE-CAST CONCRETE ELEMENTS (MANUFACTURED UNDER PLANT CONDITION)		
(a)	WALLS IN WATER RETAINING STRUCTURES	75	EXPOSED TO EARTH, WEATHER OR WATER
(b)	BEAMS, WALL & UNDERSIDES/SIDES OF SLABS OF WATER RETAINING STRUCTURES	75	IN CONTACT WITH SEA OR ABOVE SEA WATER
(c)	TOP SIDE OF SLABS (WITH FINISHED FORM SURFACES)	50	ABOVE SEA WATER

5.7 CONCRETE COVER SHALL NOT EXCEED 100mm TO AVOID SERIOUS SHRINKAGE CRACKS.

5.8 MINIMUM STANDARD 90° HOOK LENGTHS FOR REINFORCING BARS SHALL BE AS FOLLOWS:

BAR DIA	HOOK LENGTH	BAR DIA	HOOK LENGTH
12	200	28	350
16	200	32	400
20	250	36	440
24	300		

5.9 REINFORCEMENT SHALL BE SUPPORTED BY:

-PROPRIETARY CONCRETE SPACERS FOR FOOTINGS, GROUND SLABS AND FOUNDATIONS.

-STEEL/PLASTIC BAR CHAIRS FOR SUSPENDED SLABS, SIDE COVERS ON PEDESTAL FOOTINGS, BASES OR ANY VERTICAL SURFACES.

-BENT DEFORMED BAR SPACERS BETWEEN THE TOP AND BOTTOM REINFORCEMENT MATS.

6.0 ANCHOR BOLTS

6.1 ANCHOR BOLT SPECIFICATION SHALL BE IN ACCORDANCE WITH MD-101-SMEM-EG-CV-SPC-0002 AND SHALL BE OF ROUND BAR MATERIAL MINIMUM GRADE 36 STEEL TO ASTM F1554

6.2 ANCHOR BOLTS, NUTS AND WASHERS SHALL BE HOT DIPPED GALVANIZED UNLESS FABRICATED FROM STAINLESS STEEL.

6.3 PROPRIETARY ANCHORS SHALL BE M20 CHEMICAL ANCHORS INSTALLED STRICTLY IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS AND COMPLY WITH ANCHOR BOLT SPECIFICATION MD-101-SMEM-EG-CV-SPC-0002.

6.4 ALL ANCHOR BOLTS SHALL HAVE ONE WASHER AND ONE NUT UNLESS NOTED OTHERWISE ON THE DESIGN DRAWINGS.

6.5 RESIN GROUTING OF ANCHOR BOLTS AND/OR DOWEL BARS INTO PREFORMED OR DRILLED HOLES SHALL BE CARRIED OUT USING AN EMPLOYER APPROVED INJECTION SYSTEM INSTALLED STRICTLY IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.

6.6 ALL FABRICATED ANCHOR BOLTS SHALL BE TIGHTENED TO THE SNUG TIGHT CONDITION. ALL PROPRIETARY ANCHOR BOLTS SHALL BE TIGHTENED TO THE MANUFACTURER'S RECOMMENDED TORQUE.

7.0 PENETRATIONS AND CAST IN ITEMS

7.1 ALL EMBEDDED ITEMS SHALL BE IN ACCORDANCE WITH STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0006.

7.2 ALL EMBEDDED STEELWORK OTHER THAN REINFORCEMENT SHALL BE GRADE 250 OR 300 AND SHALL BE HOT DIPPED GALVANIZED.

7.3 ALL CAST-IN PLACE STEELWORK SHALL BE FIRMLY SECURED TO FORMWORK SUCH THAT ALL EXPOSED STEEL SURFACES REMAIN IN PLACE DURING THE CONCRETE POUR.

7.4 BEFORE POURING CONCRETE THE CONTRACTOR SHALL VERIFY REQUIREMENTS FOR ALL RELEVANT SERVICES, EMBEDDED ITEMS, BOLTS, SLEEVES, ETC.

7.5 NO HOLES, CHASES OR EMBEDMENTS OTHER THAN THOSE SHOWN ON THE DRAWINGS SHALL BE MADE IN THE CONCRETE WITHOUT PRIOR APPROVAL OF THE EMPLOYER.

7.6 ALL SLABS SHALL HAVE FULLY ANCHORED RE-ENTRANT BARS AT ALL RE-ENTRANT CORNERS AS SHOWN ON THE DESIGN DRAWINGS OR STANDARD DRAWING MD-101-SMEM-EG-ST-DST-0007.

8.0 EARTHING

8.1 FOR EARTHING REQUIREMENTS REFER TO ELECTRICAL DESIGN DRAWINGS AND ELECTRICAL STANDARD DETAILS OR DRAWING MD-101-SMEM-EG-EL-DST-0001.

9.0 PROTECTIVE COATING

9.1 PROTECTIVE COATING ON CONCRETE SURFACES AS NOMINATED ON DESIGN DRAWINGS SHALL BE IN ACCORDANCE WITH SPECIFICATION MD-101-SMEM-EG-ST-SPC-0009.

NOTES:

1. THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH STEELWORK GENERAL NOTES ON DRAWING MD-101-SMEM-EG-ST-DST-0009.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD

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REV C01 PREPARED FOR MA'ADEN BY HATCH ASSOCIATES H332701-0000-35-041-0001_C

MA'ADEN معادن Saudi Arabian Mining Company

DRAWN	J BADAL	01.03.11
CHECKED	J EDWARDS	31.03.11
ENGINEER	D KIRKWOOD	31.03.11
APPROVED	R COVELLA	11.05.11
MA'ADEN		
SCALE NTS		A1

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
CONCRETE GENERAL NOTES

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0001 REV C01

$FILE$ $TIME$ $DATE$